

Neighborhood Park Preliminary Design Masterplan

The plan at scale. Every room struck off the crescent's centre; every area the measured area of the drawn polygon.

02

OF 12 UPLOAD SLOTS

15,000 m²

SITE AREA

144 m

CRESCENT ARC LENGTH

2,595 m²

CANOPY AREA

131

TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL

wasim437.github.io/AI_SAFA/

01

WHAT IS IN THIS FILE

Phase5 Masterplan Development Report

PLAN

Fig10 masterplan

PLAN

Planting

DRAWING

02

THE SCALE OF THE ANALYSIS BEHIND IT

39 years

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours

of sun position ray-traced against 131 trees for every hour of the year

15,000

one-metre ground cells modelled for shade and comfort, site-wide

4 models

trained and tested for leakage — two of them decide what this document reports

41 checks

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

03

THE PHASES BEHIND THIS FILE

P5 Masterplan Development

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

Produced by **CODE** [src/plan.py](#) [src/figures.py](#) **DATA** [data/raw/site_zoning_schedule.csv](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

02

OF 12

01 THE TEN PHASES

P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

P5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

THIS DOCUMENT

P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

P4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

P6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

P8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

Plan

src

Figures

src

Site zoning schedule

data/raw

03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 02 OF 12 · PRELIMINARY DESIGN MASTERPLAN

Preliminary Design Masterplan

The room schedule, and how 15,000 m² is allocated

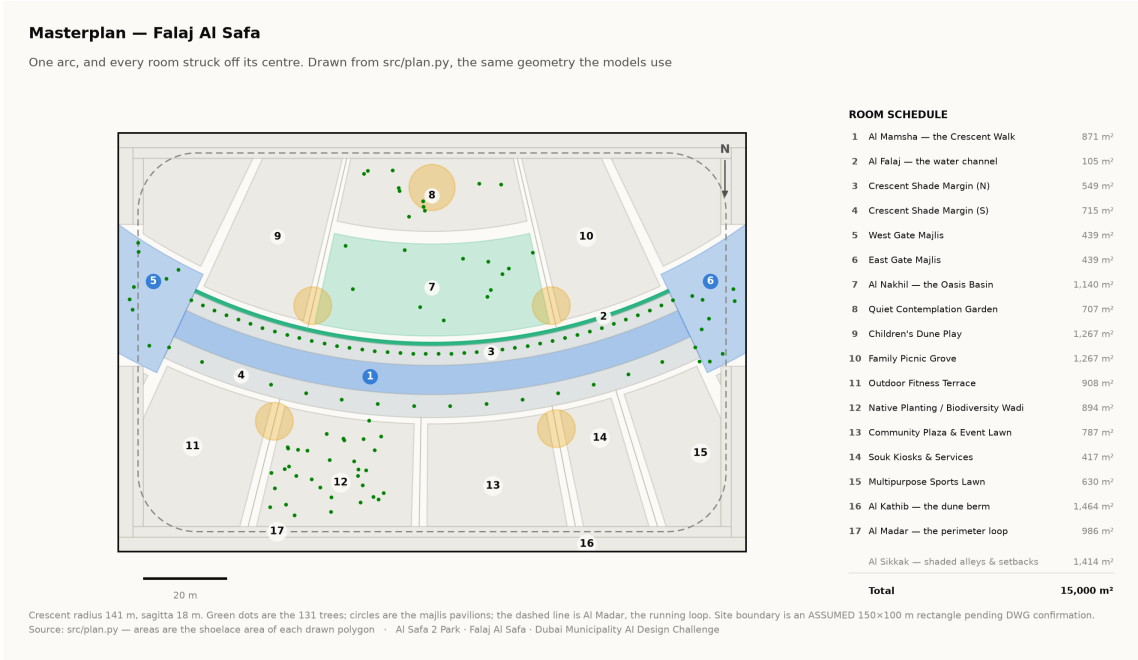
The masterplan is generated, not drawn. Every room is a box in the crescent's own polar frame, mapped to site metres and clipped to the boundary — so each area is the measured area of the polygon actually drawn, and the schedule closes on 15,000 m² without anyone reconciling a spreadsheet.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by python `run_analysis.py`

1. The organising geometry

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

The arc has a chord of 138 m and a sagitta of 18 m, giving a radius of 141.25 m and a true arc length of 144.2 m. Those four numbers generate everything else in this report.



Masterplan with the numbered room schedule. Technical drawing — to scale.

2. Room schedule

Room	Category	Area (m ²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%

Room	Category	Area (m²)	% of site
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

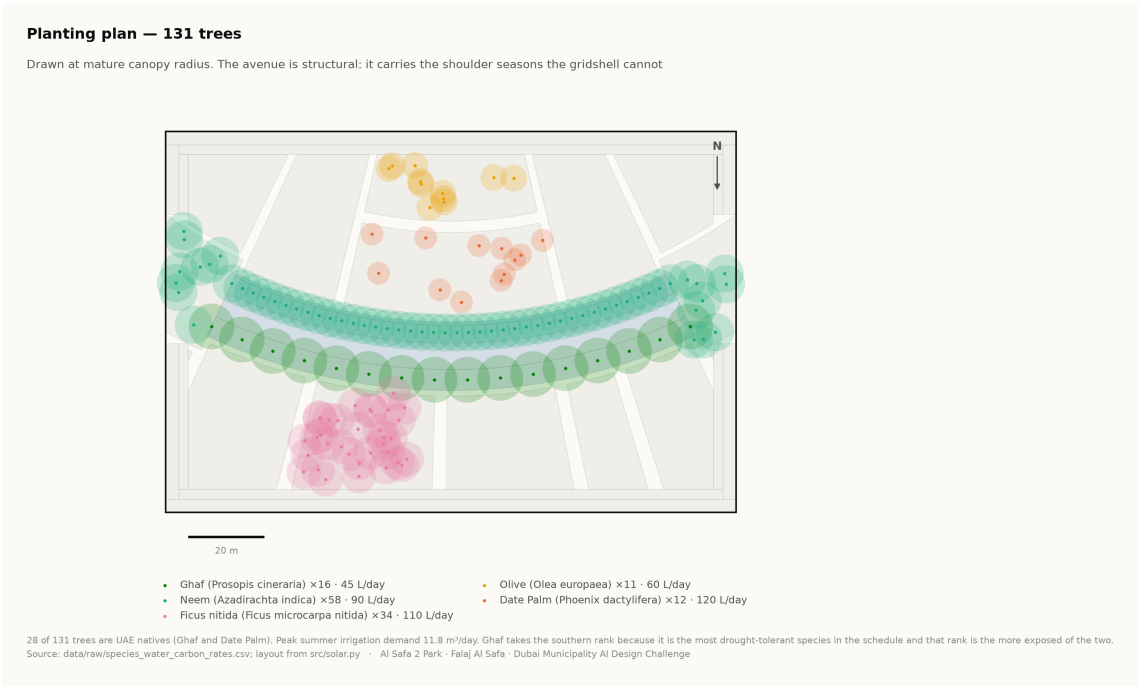
Total 15,000 m². Areas are shoelace areas of the drawn polygons, and the pipeline asserts that the schedule sums to the site area.

3. Where things are, and why

- **The hollow holds the lingering.** The Oasis Basin, the quiet garden, the children's play and the picnic grove all sit on the concave side, which the comfort model shows is measurably cooler than the convex face.
- **The convex face takes the active and the commercial.** The sports lawn, the community plaza and the souk sit south of the arc, where exposure matters less and evening use dominates.
- **The alleys are radii.** Every sikka is a radius of the same circle, so every room presents its face to the crescent rather than a corner.
- **The perimeter is a berm, not a fence.** Al Kathib is planted earth against the roads, doing three jobs at once — noise, glare and heat.

4. Site boundary — a stated assumption

The site is modelled as a 150 × 100 m rectangle. This is an **assumption** pending confirmation against the CAD file issued with the competition documents. Every area figure in this submission depends on it, and it is flagged here rather than left for a juror to discover.



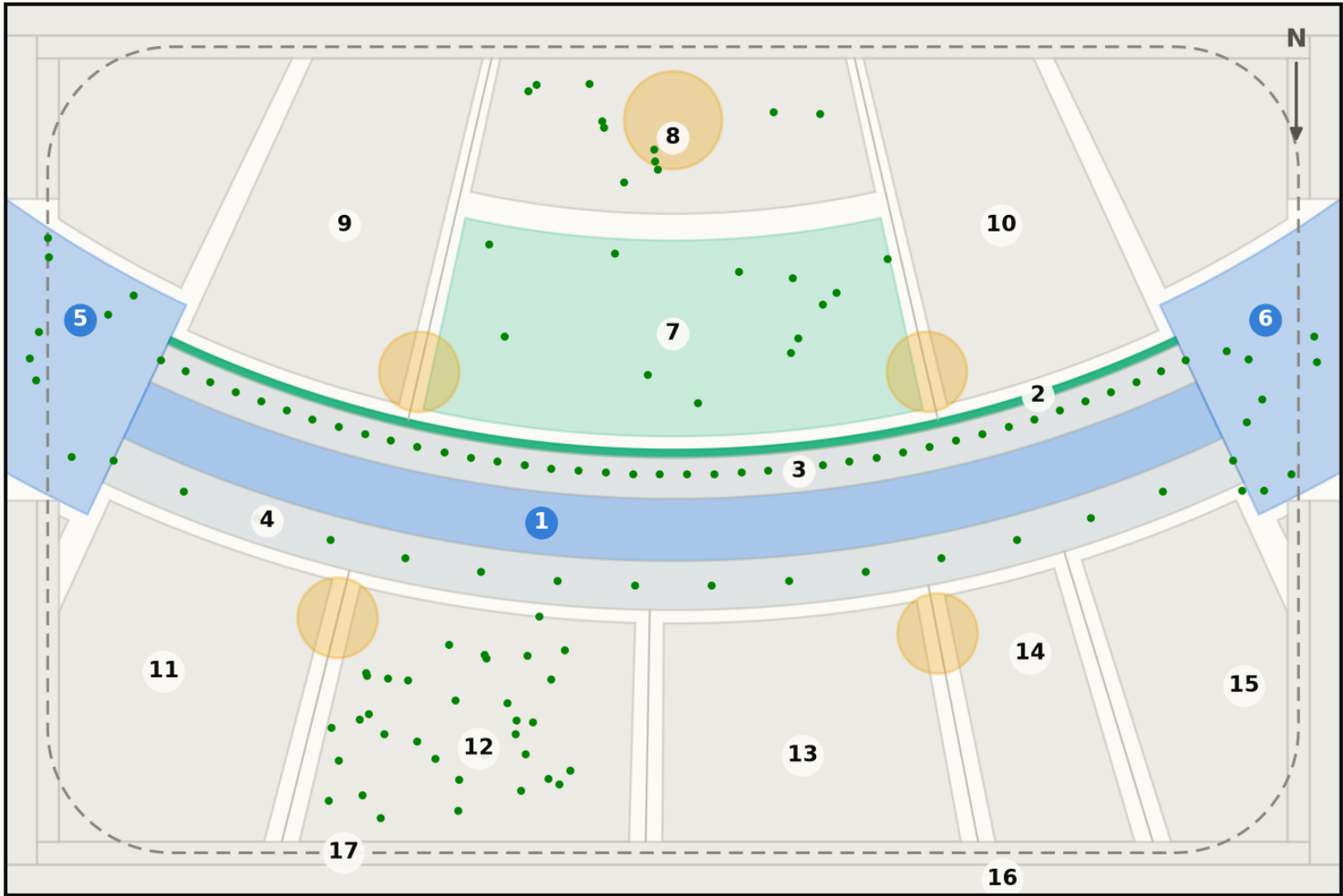
Planting plan — 131 trees at mature canopy radius. Technical drawing — to scale.

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²

Total 15,000 m²

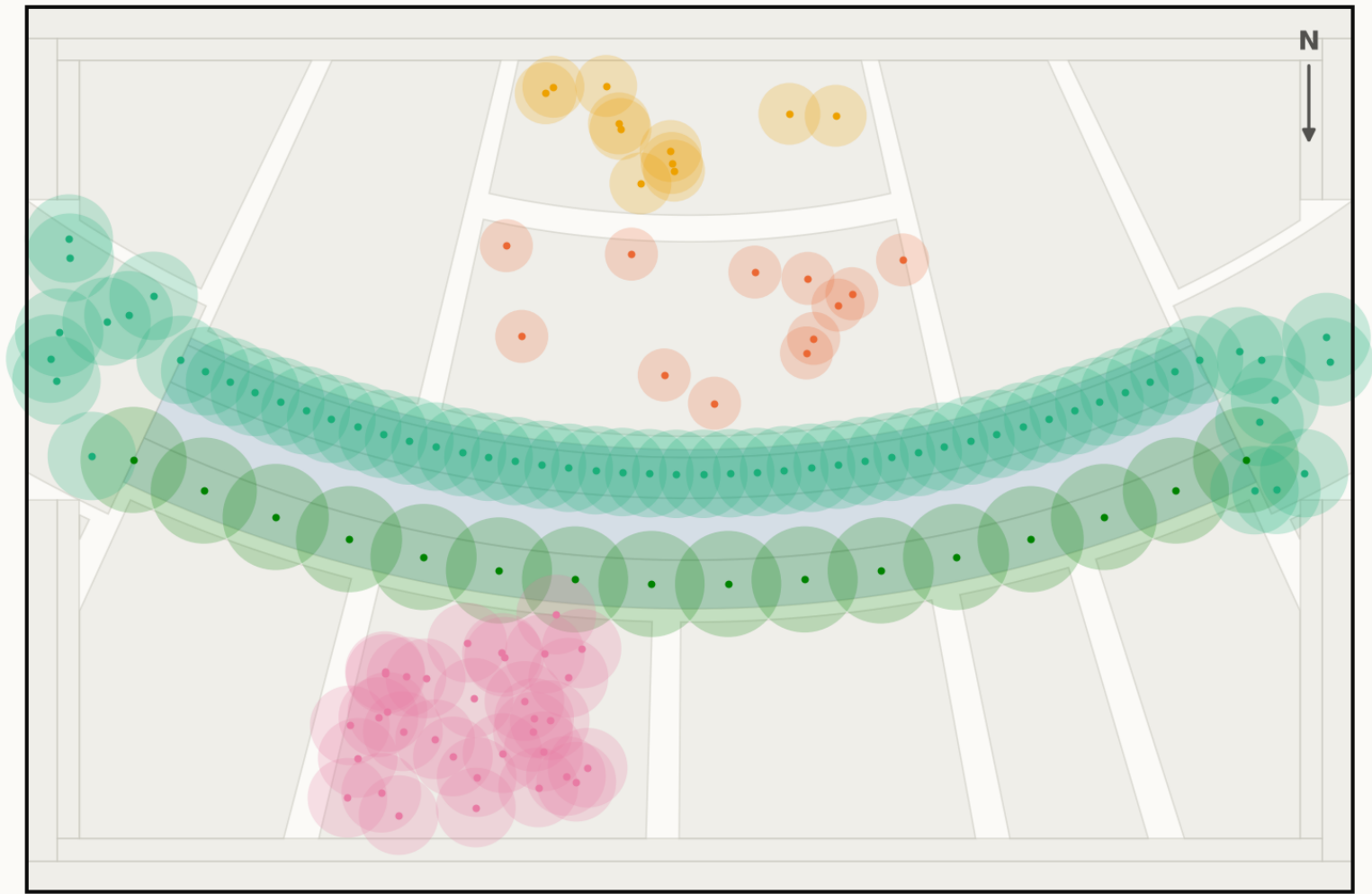
Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Planting

Technical drawing — to scale.

Planting plan — 131 trees

Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot



- Ghaf (*Prosopis cineraria*) ×16 · 45 L/day
- Olive (*Olea europaea*) ×11 · 60 L/day
- Neem (*Azadirachta indica*) ×58 · 90 L/day
- Date Palm (*Phoenix dactylifera*) ×12 · 120 L/day
- Ficus nitida (*Ficus microcarpa nitida*) ×34 · 110 L/day

28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.
Source: data/raw/species_water_carbon_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Every step of the work, with its proof attached

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

PHASE 5 Masterplan Development

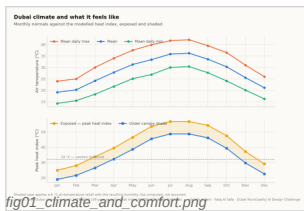
THIS DOCUMENT RESTS ON



Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

CODE [src/plan.py](#) [src/figures.py](#)

PROOF — click to open [data/raw/site_zoning_schedule.csv](#)
[data/processed/masterplan_geometry.json](#)

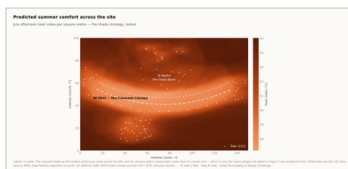


PHASE 1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

CODE [src/climate.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/hourly_climate_comfort_8760.csv](#) [data/raw/sources.json](#)

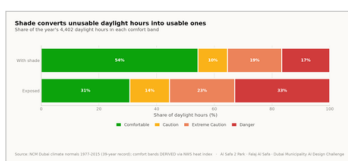


PHASE 2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

CODE [src/climate.py](#)

PROOF — click to open [models/headline_metrics.json](#)

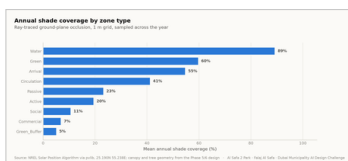


PHASE 3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

CODE [src/config.py](#) [src/plan.py](#)

PROOF — click to open [models/headline_metrics.json](#)



PHASE 4 Concept Development

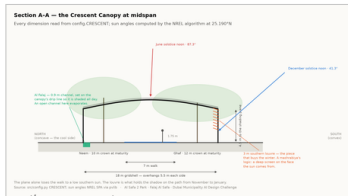
Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

CODE [src/plan.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/masterplan_geometry.json](#)

The work, with its proof attached — continued

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.



section_crescent.png

PHASE 6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

CODE [src/drawings.py](#) [src/config.py](#)

PROOF — [click to open](#) [data/processed/planting_layout.csv](#)

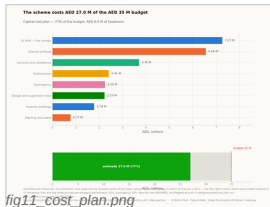


fig11_cost_plan.png

PHASE 7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

CODE [src/costing.py](#) [src/climate.py](#)

PROOF — [click to open](#) [data/processed/cost_plan.csv](#) [models/cost_summary.json](#)

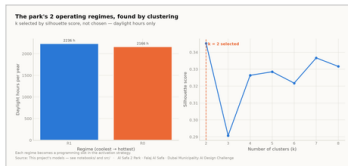


fig08_microclimate_regimes.png

PHASE 8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

CODE [src/models.py](#) [src/dataset.py](#)

PROOF — [click to open](#) [data/processed/spatial_grid_comfort.csv](#)

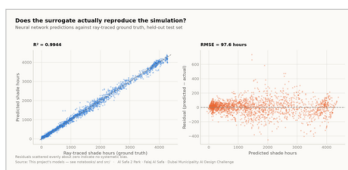


fig05_surrogate_performance.png

PHASE 9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

CODE [src/models.py](#) [src/boards.py](#) [tools/sync_film.py](#)

PROOF — [click to open](#) [models/model_metrics.json](#) [tests/test_pipeline.py](#)



board_2_evidence.png

PHASE 10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

CODE [tools/build_submission_pdfs.py](#) [tools/build_reports.py](#)

PROOF — [click to open](#) [tests/test_pipeline.py](#)

The 2 images in this file, and the whole record

This slot's own pictures come first, larger. The remaining 16 of the 18 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

01

IN THIS FILE

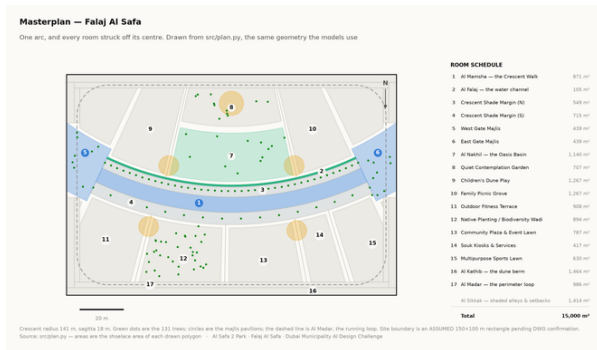
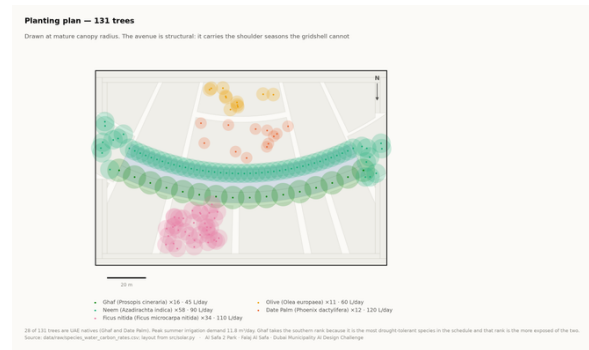


Fig10 masterplan

analysis output



Planting

technical drawing

02

THE REST OF THE RECORD

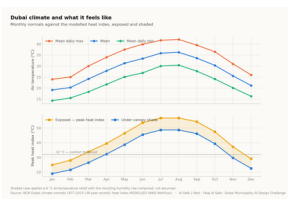


Fig01 climate and comfort

analysis output

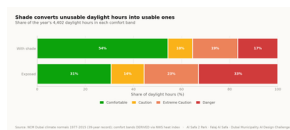


Fig02 comfort bands

analysis output

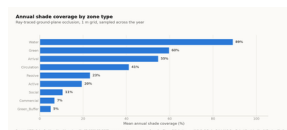


Fig03 shade by zone

analysis output

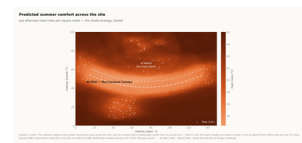


Fig04 site comfort map

analysis output

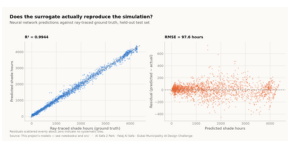


Fig05 surrogate performance

analysis output

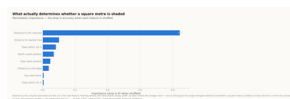


Fig06 feature importance

analysis output

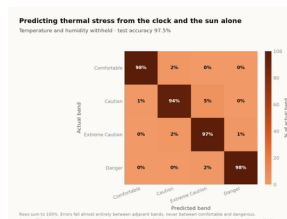


Fig07 confusion matrix

analysis output

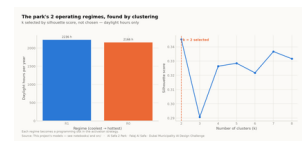


Fig08 microclimate regimes

analysis output

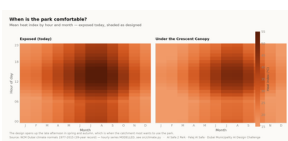


Fig09 diurnal comfort

analysis output

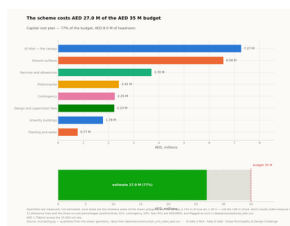
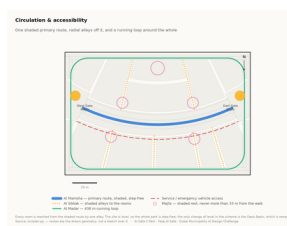


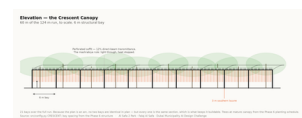
Fig11 cost plan

analysis output



Circulation

technical drawing



Elevation

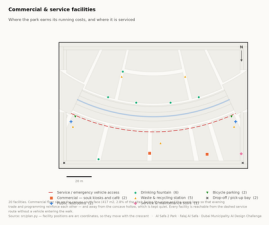
technical drawing

The visual record — continued

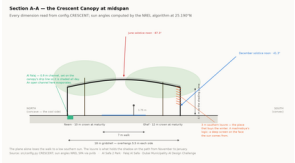
This slot's own pictures come first, larger. The remaining 16 of the 18 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

02

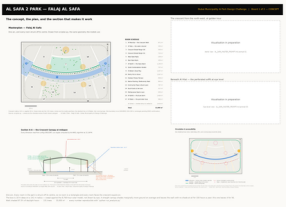
OF 12



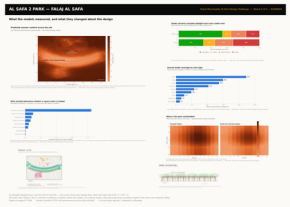
Facilities
technical drawing



Section
technical drawing



Board 1 concept
presentation board



Board 2 evidence
presentation board

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

02

OF 12

THE FIGURES THIS FILE LEADS ON

each one appears in the full tables below, from the same file

15,000 m²

SITE AREA

144 m

CRESCENT ARC LENGTH

2,595 m²

CANOPY AREA

131

TREES PLANTED

MEASURED RESULT

models/headline_metrics.json

Annual daylight hours modelled	4,402
Comfortable daylight hours — today	44.5%
Comfortable daylight hours — as designed	64.6%
Mean heat-index reduction under canopy	7.13 °C
Peak heat index, exposed	56.8 °C
Peak heat index, shaded	48.7 °C
Crescent Walk shaded, canopy and louvre	87.3%
Site-wide mean shade	34.1%
Trees planted	131

TRAINED MODELS, ON HELD-OUT TEST SETS

models/model_metrics.json

M1a Random Forest — shade surrogate, test R ²	0.9982
M1b Neural network — deployed surrogate, test R ²	0.9944
M2 Gradient Boosting — comfort band, test accuracy	97.5%
M3 K-Means — regimes selected by silhouette	k = 2
Training / validation / test split	10,500 / 2,250 / 2,250
Random seed, fixed for reproducibility	42

WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

Climate fidelity

the modelled year reproduces the published normals it was built from

No overlaps

no room in the schedule overlaps another

Areas close

every area sums back to 15,000 m²

Budget held

the cost plan stays inside AED 35 M

No leakage

no model can see its own answer in its inputs

RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js   # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The complete project, and where to check it

Every one of the 121 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE SOURCES BEHIND THIS FILE

open these first if you want to check this document

src/plan.py	Technical drawing — to scale.
src/figures.py	Analysis output — computed from project data.
data/raw/site_zoning_schedule.csv	Technical drawing — to scale.

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Ma...
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Di...
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (7)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

[figures/fig04_site_comfort_map.png](#)
[figures/fig05_surrogate_performance.png](#)
[figures/fig06_feature_importance.png](#)
[figures/fig07_confusion_matrix.png](#)
[figures/fig08_microclimate_regimes.png](#)
[figures/fig09_diurnal_comfort.png](#)
[figures/fig10_masterplan.png](#)
[figures/fig11_cost_plan.png](#)

Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation_crescent.png](#)
[design/visuals/elevation_crescent.png](#)
[design/visuals/facilities_crescent.png](#)
[design/visuals/planting_crescent.png](#)
[design/visuals/section_crescent.png](#)
[design/boards/board_1_concept.png](#)
[design/boards/board_2_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/__init__.py](#)
[src/boards.py](#)
[src/climate.py](#)
[src/config.py](#)
[src/costing.py](#)
[src/dataset.py](#)
[src/drawings.py](#)
[src/figures.py](#)
[src/models.py](#)
[src/plan.py](#)
[src/solar.py](#)
[src/viz.py](#)
[run_analysis.py](#)

Analysis module
The two presentation boards
The 8,760-hour year rebuilt from NCM normals
Every constant the project reads
The cost model against the AED 35 M ceiling
Assembles the ML training tables
Section, elevation, circulation, planting, facilities
The analysis charts
The four models
SINGLE SOURCE of the crescent geometry
Sun position and shadow ray-tracing
Analysis module
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (12)

[tools/__init__.py](#)
[tools/build_docs.py](#)
[tools/build_links.py](#)
[tools/build_notebook.py](#)
[tools/build_reports.py](#)
[tools/build_submission_pdfs.py](#)
[tools/embed_narration.py](#)
[tools/organize_repo.py](#)
[tools/report_content.py](#)
[tools/sync_film.py](#)
[tools/sync_portal.py](#)
[tools/sync_submission.py](#)

THE DATA, AS ISSUED (7)

[data/raw/construction_unit_rates_aed.csv](#)
[data/raw/dubai_climate_normals_ncm.csv](#)
[data/raw/dubai_population_dsc_2023.csv](#)
[data/raw/site_zoning_schedule.csv](#)
[data/raw/sources.json](#)
[data/raw/species_water_carbon_rates.csv](#)

Source dataset
Source dataset
Source dataset
The measured room schedule
Every source dataset, with its period
Source dataset

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The complete project — continued

[data/raw/walk_catchment_rings.csv](#)

Source dataset

THE DATA, AS PROCESSED (6)

[data/processed/cost_plan.csv](#)

The capital cost plan, line by line

[data/processed/hourly_climate_comfort_8760.csv](#)

The 8,760-hour climate and comfort series

[data/processed/masterplan_geometry.json](#)

The plan geometry, as exported

[data/processed/planting_layout.csv](#)

Every one of the 131 trees

[data/processed/schema.json](#)

Generated by the pipeline

[data/processed/spatial_grid_comfort.csv](#)

The 15,000-cell ground grid

THE TRAINED MODELS (3)

[models/cost_summary.json](#)

Model output

[models/headline_metrics.json](#)

The headline numbers

[models/model_metrics.json](#)

Trained-model metrics

THE TESTS (2)

[tests/test_pipeline.py](#)

41 correctness checks

[tests/test_film.js](#)

Every frame of the concept film

THE NOTEBOOK (1)

[notebooks/AL_SAFa_2_PARK_COMPLETE_ANALYSIS.ipynb](#)

The complete analysis, outputs embedded

THE WEBSITE AND ANALYTICS PORTAL (9)

[docs/index.html](#)

The project website

[docs/SUBMISSION_GUIDE.md](#)

Published site

[docs/_PORTAL/gallery_data.js](#)

Published site

[docs/_PORTAL/plan_geometry.js](#)

Published site

[docs/_PORTAL/portal.js](#)

Published site

[docs/_PORTAL/portal_data.js](#)

Published site

[docs/_PORTAL/selftest.js](#)

Published site

[docs/_PORTAL/DATA_AUDIT.md](#)

Published site

[docs/_PORTAL/README.md](#)

Published site

THE COMPETITION BRIEF, AS ISSUED (7)

[00_BRIEF/Ai Park - Master Plan \(4\).jpg](#)

Dubai Municipality source document

[00_BRIEF/Ai Park Scope of Work & General Terms & Conditions \(5\).pdf](#)

Dubai Municipality source document

[00_BRIEF/Ai Safa Park 2 Plan \(5\).dwg](#)

Dubai Municipality source document

[00_BRIEF/MAIN WEBSITE DETAILS.txt](#)

Dubai Municipality source document

[00_BRIEF/Neighborhood Parks Manual \(4\).pdf](#)

Dubai Municipality source document

[00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt](#)

Dubai Municipality source document

[00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt](#)

Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

[submission/01_Design_Narrative_Concept/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/02_Preliminary_Design_Masterplan/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/04_Key_Sections_Elevations/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/05_3D_Spatial_Visualizations/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/06_AI_Methodology_Report/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/07_User_Experience_Activation_Strategy/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/08_Sustainability_Concept_Strategy/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/09_Material_Landscape_Palette/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/10_Complete_Design_Report/MANIFEST.md](#)

What is in the slot, and what produced each file

[submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md](#)

What is in the slot, and what produced each file

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The complete project — continued

[submission/12_Concept_Animation_Video/MANIFEST.md](#)

What is in the slot, and what produced each file

WORKING HISTORY (1)

[archive/withdrawn_visuals/README.md](#)

Images withdrawn on purpose, and why

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